2024

AlfredCamera Moc
Doubling Developı
Foundation for Ge

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featuring Amazon Bedrock.

Overview | Opportunity | Solution | Outcome | AWS Services Used

100%

increase in software development speed

80%

reduction in infrastructure management time

20%

reduction in cloud costs

Seconds

instead of minutes to scale services

Overview

Global surveillance software company AlfredCamera wanted to modernize the monolithic design of the application behind its home security camera app. The company, which launched its AlfredCamera app on Amazon Web Services (AWS), rearchitected the application for microservices, with Amazon Elastic Kubernetes Service (Amazon EKS) simplifying the task of managing cloud-based Kubernetes and AWS Lambda providing computing services.

By modernizing its application on AWS, AlfredCamera doubled its speed of development while reducing management time by 80 percent, lowered costs by 20 percent, and scales in seconds. Plus, AlfredCamera lays the foundation for generative AI adoption, testing its use cases with Amazon Bedrock.

Opportunity | Targeting the Growing Home Security Market Worldwide

AlfredCamera is a home security camera app that turns spare smartphones into powerful security cameras. Launched in 2014, the app has accumulated more than one hundred million downloads across 175 regions and gained awards such as 'Most Popular Utility App of 2019' on the Play Store. In 2022, the company launched AlfredCamera, a security camera device that aims to enhance user experiences and ensure comprehensive peace of mind through integrated hardware and software services.

As the home security industry expands globally, the company sees millions more potential customers worldwide seeking an easy-to-use home camera solution. Alex Song, the founder and CEO of AlfredCamera, says, "There's more than 6 billion smartphone users around the world today. That's over 80 percent of the global population. By recreating essential and advanced security camera features in an app, we democratized DIY home security."

From the beginning, AlfredCamera has built its app on Amazon Web Services (AWS), using core services such as Amazon Elastic Compute Cloud (Amazon EC2), Elastic Load Balancing (ELB), and Amazon Simple Storage Service (Amazon S3) for petabytes of storage. However, to speed up innovation, the company decided to transition its app's underlying architecture from monolithic to microservices. Giga Sun, senior backend engineer at AlfredCamera, says, "Product development is key as competition in our market expands. We knew that microservices would make it easier to maintain a high pace of development and stay ahead."

Solution | Doubling Development Speed with Microservices on AWS

AlfredCamera rebuilt its application to run in a microservices architecture with Amazon Elastic Kubernetes Service (Amazon EKS), which runs Kubernetes in the cloud. "Amazon EKS offered seamless integration with other AWS services, providing a significant advantage," says Sun "Plus, we didn't have to build the whole Kubernetes service from scratch—which saved at least a couple of months of development time."

The business also uses AWS Lambda, a serverless computing service which performs several t checking the performance of endpoints. In addition, the company migrated from Redis to Am which delivers real-time, cost-optimized performance for modern applications.

To stream camera imagery, AlfredCamera is using Amazon CloudFront as its content delivery network. It also continues to use Amazon S3, deployed across 10 AWS Regions globally, ensuring rapid customer data retrieval within milliseconds (ms); and Amazon S3 Storage Lens, which provides visibility into activity trends. Amazon S3 Storage Lens complements the interactive query service Amazon Athena, enhancing data analytics capabilities for more informed, data-driven decision-making.

By moving to a microservices architecture on AWS, AlfredCamera can innovate faster—doubling its development speed. It also reduced costs by 20 percent by using a Kubernetes plug-in to schedule Amazon EC2 Spot Instances to reduce its reliance on Amazon EC2 Reserved Instances. In addition, infrastructure scaling is now completed in seconds rather than minutes, latency has fallen to below 80ms, and downtime has been eliminated. Adds Sun, “We’ve cut management time by at least 80 percent thanks to our integration between Amazon EKS and GitHub, largely automating deployments.”

Outcome | Setting the Foundation for Generative AI and Supporting Millions of Customers

With faster innovation powered by microservices on AWS, AlfredCamera can consistently drive growth in the expanding surveillance market. The company can effortlessly expand its infrastructure globally, while ensuring reliability and security—which are non-negotiable for customers.

As millions more download the app in the coming years, microservices will also streamline AlfredCamera’s ability to apply security patches and frequent software updates. “With our modernized architecture on AWS, we can stay one step ahead, focusing on customer safety and giving peace of mind to millions of users,” adds Sun.

Generative AI is transforming the home security industry, and AlfredCamera’s modernization on AWS is paving the way for the development of generative AI applications for its solution. Generative AI enables new face recognition, object detection, and activity recognition capabilities, and AlfredCamera is actively exploring use cases and discovering the best ways to apply the technology.

As part of its journey towards generative AI adoption, AlfredCamera has started testing Amazon Bedrock to support Claude 3, a vision-language model. “Our mission has always been to make home security simple, affordable, and accessible for all. By integrating generative AI across our operations and optimizing visual language models, we are automating content description and enhancing video content analysis," says Song. “Trialing Amazon Bedrock is part of our strategic approach to embracing generative AI tools. This initiative drives innovation and significantly advances our vision to build a world where everyone feels carefree, elevating the standard of service we provide to our users.”

About AlfredCamera

Headquartered in Taiwan, AlfredCamera is dedicated to simplifying home security, making it affordable and accessible to everyone. The company developed a globally used home security camera app, which has been downloaded more than 100 million times across 175 regions, receiving an average rating of 4.8 out of 5 stars from reviewers.

AWS Services Used

Amazon Elastic Kubernetes Service

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

Learn more »

AWS Lambda

Run code without thinking about servers or clusters. Only pay for what you use.

Learn more »

Amazon ElastiCache

Amazon ElastiCache is a serverless, Redis- and Memcached-compatible caching service delivering real-time, cost-optimized performance for modern applications.

Learn more »

Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies like AI21 Labs, Anthropic, Cohere, Meta, Mistral AI, Stability AI, and Amazon through a single API, along with a broad set of capabilities you need to build generative AI applications with security, privacy, and responsible AI.

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